Allocate min (2,6) = 2 units to this cell.

	D_2	D_3	a_i
O_1	(2) 2	3	2/0
O_2	3	2	8
O_3	2	2	4
b_j	6/4	8	

The reduced table is

	D_2	D_3	a_i
O_2	3	2	8
O_3	2	2	4
bj	4	8	

The least cost is 2 corresponds to the cells (O_2, D_3), (O_3, D_2), (O_3, D_3)

Allocate min (8,8) = 8 units to this cell.

	D_2	D_3	a_i
O_2	3	(8) 2	8/0
O_3	2	2	4
b_j	4	8/0	

The reduced table is

	D_2	a_i
O_3	2	4
b_j	4	

Here allocate 4 units in the cell (O_3, D_2)

	D_2	a_i
O_3	(4) 2	4/0
b_j	4/0	

Thus we have the following allocations:

	D_1	D_2	D_3	D_4	a_i
O_1	(4) 1	(2) 2	3	4	6/2/0
O_2	4	3	(8) 2	5	8/0
O_3	5	(4) 2	2	(6) 1	10/4/0
b_j	4/0	6/4/0	8/0	6/0	

Transportation schedule :

$O_1 \rightarrow D_1$, $O_1 \rightarrow D_2$, $O_2 \rightarrow D_3$, $O_3 \rightarrow D_2$, $O_3 \rightarrow D_4$

Total transportation cost

$= (4\times1)+ (2\times2)+(8\times2)+(4\times2)+(6\times1)$

$= 4+4+16+8+6$

$= ₹ 38.$

Example 10.4

Determine how much quantity should be stepped from factory to various destinations for the following transportation problem using the least cost method.

		Destination				
		C	H	K	P	Capacity
Factory	T	6	8	8	5	30
	B	5	11	9	7	40
	M	8	9	7	13	50
	Demand	35	28	32	25	

Cost are expressed in terms of rupees per unit shipped.

Solution:

Total Capacity = Total Demand

$\therefore$ The given problem is balanced transportation problem.

Hence there exists a feasible solution to the given problem.

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